Centurion IMU board (rev 1.0)

Centurion IMU board, rev 1.0. Plugs into centurion J5 with a straight JST GH 8-pin cable (pin n to pin n), 30 cm or less.

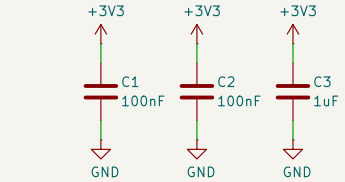
SPI mode 0 or 3, 10 MHz max. Centurion has 100 ohm series resistors on SCK, MOSI and CS (R8, R9, R41); R1 here is the matching one on MISO.

BMI270: firmware must upload Bosch's 8 kB config file after every power-up (INIT_CTRL / INIT_DATA), and do one dummy read to select SPI.

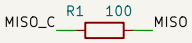
Axes as printed on the top silkscreen: +X toward J1, +Y toward H2, +Z up out of the board.

J5 pin 7 (IMU_INT1) is BMI270 INT1 (pin 4); J5 pin 8 (IMU_INT2) is BMI270 INT2 (pin 9).

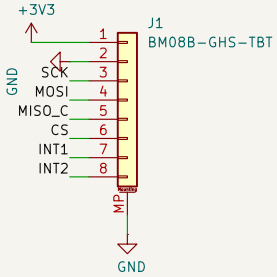
Decoupling: 100 nF at VDD and VDDIO, 1 uF bulk



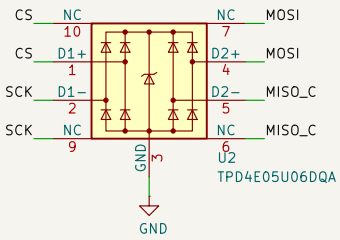
R1: MISO series resistor



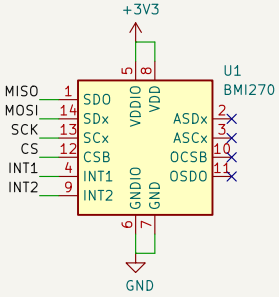
J1: to centurion J5



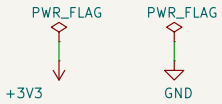
U2: ESD at the jack



U1: BMI270 IMU



Power flags: the cable from centurion J5 supplies +3V3 and GND



Mounting: two #8-32 holes, 25.4 mm (1 in) apart, not tied to GND



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UWAT VEXU

Sheet: /

File: imuBoard.kicad_sch

Title: Centurion IMU board

Size: A4	Date: 2026-11-04	**Rev: 1.0**
KiCad E.D.A. 10.0.8		Id: 1/1